

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

3th Semester B.Tech.(CE/IT)

University Exam November 2012

MA-201 Discrete Mathematics

Date: 26/11/2012

Time: 01:30 pm to 04:30 pm

Maximum Marks: 70

Instruction:

1. This question paper contains two sections **Section-I** and **Section-II**.
2. **Section-I** carries 35 and **Section-II** carries 35 marks.
3. Section wise separate answer book to be used.
4. Figure to the right indicates the distribution of full marks for the respective question.

Section-I

Q.1 Select a most appropriate option/options.

[07]

- (a) Let (N, D) , a Natural Numbers with division relation. Then (N, D) is
i) Symmetric ii) Irreflexive iii) Asymmetric iv) None of these
- (b) Which of the following is a proposition?
i) My name is Paul.
ii) Please submit your homework in time.
iii) $1+1=2$.
iv) What is your name?
- (c) $(\{2, 3, 6, 12, 24, 36, 72\}, D)$, The least upper bound of $\{2, 4, 12\}$ is
i) 24 ii) 36 iii) 72 iv) 12
- (d) Which of the following is not lattice?
i) $(R, \leq)$ ii) $(\{1, 2, 3, 6, 12, 18\}, D)$ iii) $(P(X), \subseteq)$ iv) All of these
- (e) Which of the following is distributive lattice?
i) (S_{18}, D) ii) $(S_{18}, \leq)$ iii) Both iv) None
- (f) Let $P: (N, D)$ is not lattice. and $Q: \text{Every chain is lattice.}$ then P and Q are respectively
i) True, False ii) True, True iii) False, False iv) False, True.
- (g) For $a, b \in L$, $a \leq b \leftrightarrow a * b = a \leftrightarrow a \oplus b = b$ is may not be true in
i) POSET ii) Chain iii) Lattice iv) All of these

Q.2 (a) Define (i) Lattice (ii) Principal Conjunctive Normal Form.

[04]

(b) Define Poset and give example of Poset which is not lattice.

[04]

(c) Explain distributive laws and associative laws using truth table.

[06]

OR

Q.2 (a) Define (i) Bonded Lattice (ii) Complemented Lattice.

[04]

(b) In usual notation prove that for $a, b \in L$, $a \leq b \leftrightarrow a * b = a \leftrightarrow a \oplus b = b$.

[05]

(c) Find the sum of products expansions of the following Boolean functions.

[05]

(i) $f(x, y, z) = (x + z)y$ (ii) $f(x, y, z) = 1$, if $x + z = 0$

Q.3 (a) Draw the Hasse diagram of the lattices:

[05]

(i) S_{100} (ii) S_{36}

- (b) Find Principal Disjunctive Normal Form for the expression $(\neg p \Rightarrow r) \wedge (p \Leftrightarrow q)$. [05]
- (c) In a Boolean algebra prove that $(a + b)(a' + c) = ac + a'b + bc = ac + a'b$. [04]

OR

- Q.3 (a) Verify that expression $(a' * b * c) \oplus (a * b' * c) \oplus (a * b * c')$ is symmetric or not. [05]
- (b) Write down converse, inverse and Contrapositive of the statement, If x is a number greater than 3, it can be expressed as sum of prime numbers always.. [05]
- (c) Simplify the following Boolean expression $(a * b)' \oplus (a \oplus b)'$ [04]

Section-II

- Q.4 Select a most appropriate option/options. [07]
- (a) A full m -ary tree with i internal vertices has _____ vertices.
 i) $m - i$ ii) $mi + 1$ iii) $mi - 1$ iv) $m + i + 2$
- (b) There are _____ edges in k -regular graph with n vertices.
 i) ${}^k C_2$ ii) $(k-1)C_2$ iii) $\frac{n(k-1)}{2}$ iv) $\frac{n(k+1)}{2}$
- (c) From the Figure 1 which of the following is true?
 i) Graph 1 and 2 are isomorphic. ii) Graph 1 and 3 are isomorphic.
 iii) Graph 1 and 4 are isomorphic. iv) none of these
- (d) Let $G = (V, E)$ be a graph with $|V| = n$. Then which of the following is/are true?
 i) G can not be complete binary tree with $n = 6$. ii) G can not have spanning tree.
 iii) G can be complete binary tree with $n = 6$. iv) G can have spanning tree.
- (e) Euler's formula in standard notation is _____.
 i) $n + r = e + 2$ ii) $n + e = r + 2$ iii) $n + r = e - 2$ iv) $n - r = e + 2$
- (f) Which of the following is/are true?
 i) $K_{3,3}$ is non-planar graph. ii) $K_{3,3}$ is planar graph.
 iii) K_5 is planar graph. iv) K_5 is non-planar graph.
- (g) In a simple, connected planar graph G , the relation between r regions, v vertices ($v \geq 3$) and e edges ($e \geq 1$), is _____.
 i) $e \leq 3v - 6$ ii) $e < 3v - 6$ iii) $e \geq 3v - 6$ iv) $e > 3v - 6$

- Q.5 (a) Attempt any three [12]
- (i) Draw connected graphs with 7 vertices having degree (1) 2, 4, 5, 6, 7, 2, 2, and (2) 1, 1, 1, 1, 1, 1, 2 if exists. Are these graphs bipartite?
- (ii) Define Eulerian graph and Hamiltonian graph. Draw a graph which is Hamiltonian but not Eulerian if possible.
- (iii) Draw the following graphs if possible.
 i) Strongly connected but not unilaterally connected.
 ii) Unilaterally connected but not weakly connected.
- (iv) Find $G_1 \cup G_2$, $G_1 \cap G_2$ and $G_1 \oplus G_2$ for G_1 and G_2 in Figure 2.
- (b) Write a elementary cycle of length 3 and simple path of length 6 if possible from the Figure 3. [02]

OR

- (b) Define mixed graph with illustration.

Q.6 (a) Define adjacency matrix and write a incidence and adjacency matrix form the figure **Figure 4**. [05]

(b) Apply suitable algorithm to find shortest path form s to t in **Figure 5**. [05]

(c) Define sub graph and spanning sub graph with illustration. [04]

OR

Q.6 (a) Define tree. Find the number of vertices of degree 1 in any tree having 4 vertices of degree 3, 2 vertices of degree 4 and 4 vertices of degree 2. [05]

(b) Define odd vertices. Prove that odd number of vertices are even in any undirected graph. [05]

(c) Define path and cycle with illustration. [04]

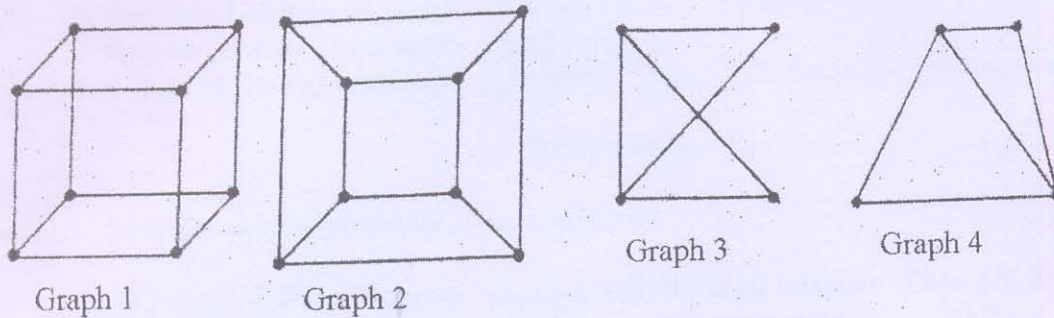


Figure 1

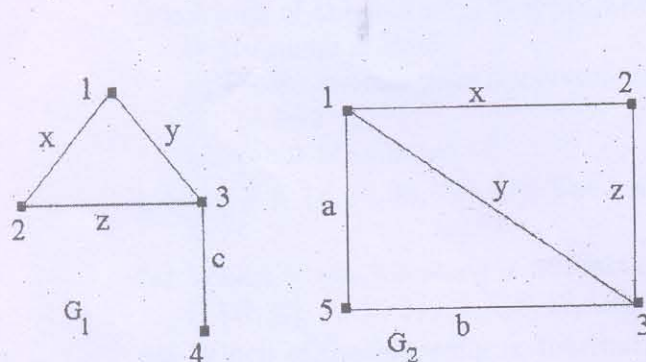


Figure 2

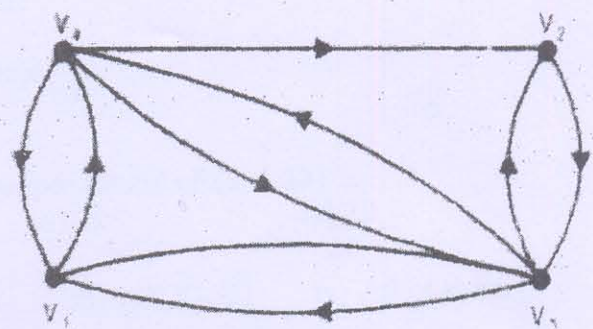


Figure 3

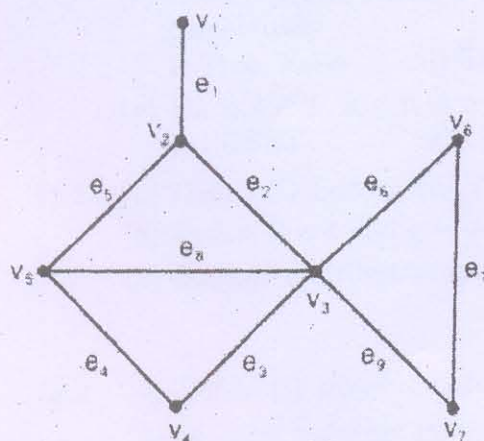


Figure 4

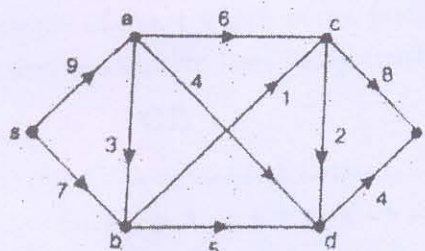


Figure 5

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